AGENTIC-ARBITER -- FREE-COOLING DECISION REPORT

Data centres over-cool, continuously, because nobody can promise them the hours ahead. A chiller plant needs hours of notice to change mode, and a thermometer only ever reports NOW -- so the mechanical chillers keep running through hours that outside air could have cooled for free. FortyGuard closes that gap with heat intelligence 2 m above the ground, the height a ground-mounted condenser actually breathes. This agent turns that forecast into an hour-by-hour schedule carrying a calibrated safety bound.

THE SITE

Location	165 Halsey / Equinix NY1 / Databank EWR1, NJ -- station KEWR, America/New_York
Building	OSM 387208881 165 Halsey / Equinix NY1 / Databank EWR1
Facade gap	not applicable -- one building, no second facade
Weather record	43,787 real hourly records from KEWR
Plume physics	NOT MODELLED. No other tagged data centre lies inside the solver's validated range, so there is no neighbour intake for a plume to arrive at: the quantity does not exist here rather than being unmeasured. This is a statement about the model's domain, not a claim that recirculation here is zero. A building's own exhaust re-entering its own intake is not modelled at any site in this project.

FortyGuard data: weather and geometry are this site's own; the four measured FortyGuard level offsets and the N-26 coverage record are Ashburn's, because only Ashburn has forecast/outcome day pairs. Quote the hours for this site; quote the coverage for Ashburn.

THIS REPORT IS A SNAPSHOT, NOT THE LIVE PAGE

The interface recomputes the decision for whatever configuration a reader selects. This file was generated at build time for the ONE configuration named below, chosen because it is the configuration in which this site's agent does the most while still changing mode at least once. If the numbers on screen differ from the numbers here, the configuration differs -- compare the two lists before concluding anything else.

Day	2025-03-29 (crossing)
Plant limit	18.0 C
Notice required	3 h
Forecast skill	0.50 relative to persistence
Level anchor	sensor
Condenser bank	longest facade
Switch budget	1 mode changes per day
Minimum dwell	3 h
Max dew point	15.0 C (Green Grid WP#46 p.6)

WHAT THE AGENT DID ON THIS DAY

The agent ran free cooling for 3 of 24 hours with 1 mode change. The reactive incumbent took 9 hours -- 6 more -- but declared 0 unsafe hours against the agent's 0. The incumbent had to break its own switch budget 1 time to stay safe; the agent never did. 9 hours were safe but still ran chillers, because the schedule could not afford them.

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Agent        3 of 24 hours free cooling, 1 mode change(s), 0 unsafe hour(s)
Incumbent    9 of 24 hours free cooling, 2 mode change(s), 0 unsafe hour(s)
           the incumbent broke its own switch budget 1 time(s) to stay safe
           9 hour(s) passed every gate and still ran chillers, because the schedule could
           not afford them
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EVERY HOUR, AND THE REASON FOR IT

hh	mode	safe	binding	bound	limit	actual	margin
00	mechanical	yes	switch budget	13.330	18.0	13.330	0.726

01	mechanical	yes	switch budget	13.055	18.0	12.780	0.674
02	mechanical	yes	switch budget	13.055	18.0	12.780	0.630
03	mechanical	yes	switch budget	13.050	18.0	12.220	0.830
04	mechanical	yes	switch budget	12.500	18.0	11.670	0.741
05	mechanical	yes	switch budget	12.780	18.0	12.220	0.640
06	mechanical	yes	switch budget	13.055	18.0	12.780	0.671
07	mechanical	yes	switch budget	14.170	18.0	13.890	0.973
08	mechanical	yes	switch budget	16.390	18.0	15.560	1.389
09	mechanical	no	dry-bulb	18.890	18.0	18.890	1.425
10	mechanical	no	dry-bulb	21.390	18.0	22.780	1.273
11	mechanical	no	dry-bulb	23.615	18.0	26.110	1.146
12	mechanical	no	dry-bulb	26.110	18.0	28.330	1.141
13	mechanical	no	dry-bulb	27.780	18.0	28.890	0.933
14	mechanical	no	dry-bulb	28.330	18.0	27.220	1.022
15	mechanical	no	dry-bulb	29.170	18.0	27.780	0.893
16	mechanical	no	dry-bulb	29.165	18.0	27.780	1.039
17	mechanical	no	dry-bulb	27.780	18.0	27.780	0.874
18	mechanical	no	dry-bulb	20.555	18.0	13.330	0.869
19	mechanical	no	dry-bulb	19.725	18.0	11.670	1.011
20	mechanical	no	dry-bulb	18.890	18.0	10.000	1.034
21	FREE	yes	-	11.665	18.0	10.000	0.942
22	FREE	yes	-	10.000	18.0	8.330	0.850
23	FREE	yes	-	8.890	18.0	7.780	0.773

bound = the agent's 90 %-nominal upper bound on intake air, forecast plus margin
actual = what the intake temperature turned out to be, from the station record
margin = group-conditional forecast error for that hour of day, plus plume spread

THE REASONING, HOUR BY HOUR

00:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 13.330 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

01:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 13.055 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

02:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 13.055 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

03:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 13.050 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

04:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 12.500 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

05:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 12.780 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

06:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 13.055 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

07:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 14.170 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

08:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 16.390 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

09:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 18.890 C against a 18.0 C limit, so it fails by 0.890 C. It would take a limit 0.890 C higher, or a bound 0.890 C tighter, to change this hour.

10:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 21.390 C against a 18.0 C limit, so it fails by 3.390 C. It would take a limit 3.390 C higher, or a bound 3.390 C tighter, to change this hour.

11:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 23.615 C against a 18.0 C limit, so it fails by 5.615 C. It would take a limit 5.615 C higher, or a bound 5.615 C tighter, to change this hour.

12:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 26.110 C against a 18.0 C limit, so it fails by 8.110 C. It would take a limit 8.110 C higher, or a bound 8.110 C tighter, to change this hour.

13:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 27.780 C against a 18.0 C limit, so it fails by 9.780 C. It would take a limit 9.780 C higher, or a bound 9.780 C tighter, to change this hour.

14:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 28.330 C against a 18.0 C limit, so it fails by 10.330 C. It would take a limit 10.330 C higher, or a bound 10.330 C tighter, to change this hour.

15:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 29.170 C against a 18.0 C limit, so it fails by 11.170 C. It would take a limit 11.170 C higher, or a bound 11.170 C tighter, to change this hour.

16:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 29.165 C against a 18.0 C limit, so it fails by 11.165 C. It would take a limit 11.165 C higher, or a bound 11.165 C tighter, to change this hour.

17:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 27.780 C against a 18.0 C limit, so it fails by 9.780 C. It would take a limit 9.780 C higher, or a bound 9.780 C tighter, to change this hour.

18:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 20.555 C against a 18.0 C limit, so it fails by 2.555 C. It would take a limit 2.555 C higher, or a bound 2.555 C tighter, to change this hour.

19:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 19.725 C against a 18.0 C limit, so it fails by 1.725 C. It would take a limit 1.725 C higher, or a bound 1.725 C tighter, to change this hour.

20:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 18.890 C against a 18.0 C limit, so it fails by 0.890 C. It would take a limit 0.890 C higher, or a bound 0.890 C tighter, to change this hour.

21:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 11.665 C, which is 6.335 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 0.942 C, and the margin is measured, not chosen: 0.942 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction differs from the one planned for, at the measured spread of wind direction over this lead time.

22:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 10.000 C, which is 8.000 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 0.850 C, and the margin is measured, not chosen: 0.850 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction differs from the one planned for, at the measured spread of wind direction over this lead time.

23:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 8.890 C, which is 9.110 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 0.773 C, and the margin is measured, not chosen: 0.773 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction differs from the one planned for, at the measured spread of wind direction over this lead time.

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